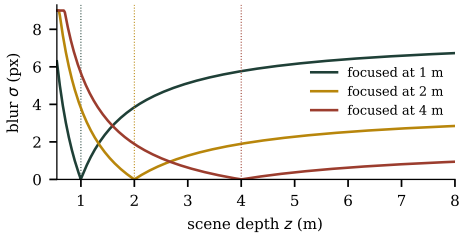
Monocular Depth Estimation: Benchmarking and a Defocus-Informed Refinement

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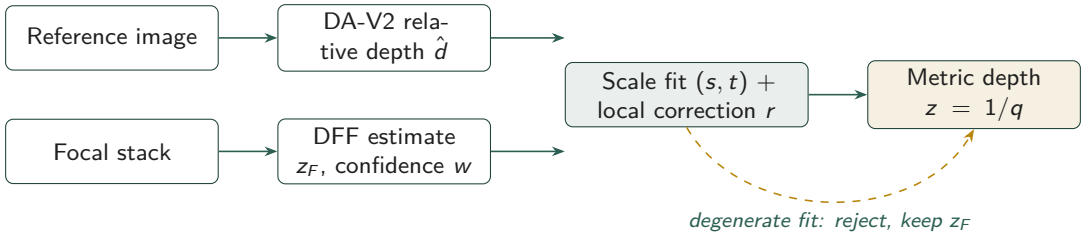
One image cannot give meters

- A small nearby object and a large far one project identically: learned depth is defined only up to scale and shift.
- Result: per-image alignment is required before a relative model can report meters.
- Defocus is different: blur is set by thin-lens geometry, so a focal stack carries an absolute scale cue.



Blur σ vs. scene depth for three focus settings. The curve is anchored in meters, not relative units.

Three components, one depth map

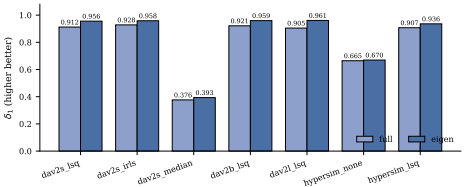


- **Learned branch:** Depth Anything V2 (ViT-S/B/L + metric indoor checkpoint), evaluated on 654 NYU-Depth v2 test images.
- **Classical branch:** depth from focus on rendered stacks (6 scenes, tune/eval split) and DDFF 12-Scene (120 real stacks).
- **Ours:** focus-informed scale fitting and regularized refinement, with rejection and a focus fallback.

Learned baseline: alignment is the whole game

Checkpoint	Alignment	AbsRel	RMSE (m)	δ_1
ViT-S	median scale	0.494	2.349	0.393
ViT-S	least squares	0.062	0.297	0.956
ViT-S	robust IRLS	0.058	0.292	0.958
ViT-B	least squares	0.057	0.283	0.959
ViT-L	least squares	**0.055**	**0.277**	**0.961**
Metric-Indoor-S	none (raw)	0.218	0.650	0.670
Metric-Indoor-S	least squares	0.077	0.353	0.936

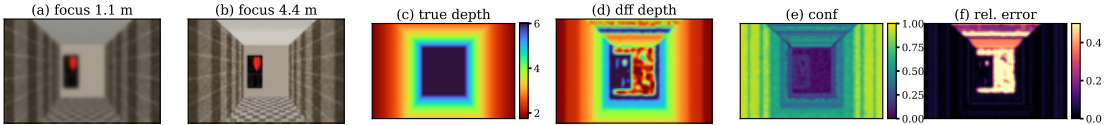
NYU-Depth v2 test split, 654 images, Eigen crop.



δ_1 by checkpoint, full vs. Eigen crop.

- Median scale collapses (0.494): a scalar cannot fix inverse-depth offset. Affine $q = s\hat{d} + t$ is the minimum.
- The metric checkpoint has a scale prior but is biased on NYU: 0.218 raw vs. 0.077 aligned.
- S $\rightarrow$ L buys $\approx$0.005 δ_1 at $\sim$4$\times$ runtime: ViT-S is our refinement backbone.

Depth from focus on synthetic stacks



Corridor: two focus slices, ground-truth depth, DFF estimate, confidence, relative error.

- Modified Laplacian focus measure per frame; parabolic fit around the peak gives sub-frame z_F.
- Confidence $w \in [0, 1]$ marks flat or ambiguous focus curves; $\bar{w} \approx 0.06$–0.26 textureless vs. 0.26–0.66 textured.
- All-focus guided smoothing in inverse depth recovers flat regions (textureless AbsRel $0.77 \rightarrow 0.61$ held out).
- Accurate where texture exists; degrades on flat surfaces and at occlusion boundaries (defocus veiling).

Contribution: defocus-informed scale fitting

Affine fit on inverse depth plus a local correction r:

$$q(\mathbf{x}) = s\,\hat{d}(\mathbf{x}) + t + r(\mathbf{x}), \qquad z(\mathbf{x}) = q(\mathbf{x})^{-1}$$

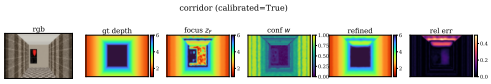
$$\min_{s,t,r} \sum_{\mathbf{x}} w(\mathbf{x})\,\rho\Big(q(\mathbf{x}) - z_F(\mathbf{x})^{-1}\Big) + \lambda \sum_{\mathbf{x}} \|\nabla r(\mathbf{x})\|^2 + \mu \sum_{\mathbf{x}} r(\mathbf{x})^2, \qquad q \geq \epsilon > 0$$

- Huber loss ρ, confidence-weighted: only reliable focus pixels pull the fit.
- Affine (s, t) first with $r = 0$, then local corrections; λ, μ suppress large or abrupt edits where w is weak.
- **Rejection:** anticorrelated predictions make the fit degenerate ($s \leq 0$); the output is marked uncalibrated and falls back to z_F instead of emitting bad metric depth.

Held-out synthetic scenes: combined beats both parts

Method	AbsRel	RMSE (m)	δ_1
DFF raw	0.233	1.552	0.732
DFF + all-focus smoothing	0.221	1.510	0.702
DA-V2 aligned (oracle)	0.271	1.524	0.548
DA-V2 + scale fit	0.308	1.614	0.459
Refined (ours)	0.259	1.483	0.575
Combined (ours)	**0.206**	**1.284**	0.704

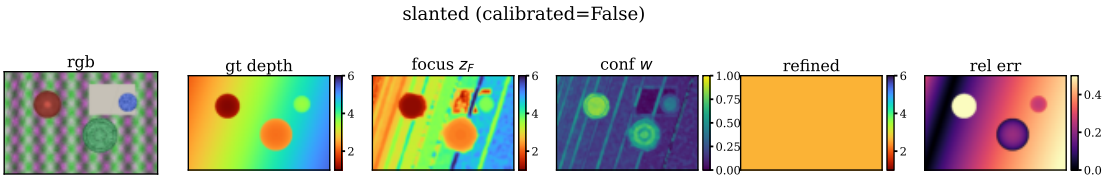
Mean over 4 held-out scenes (corridor, objects, slanted, shelves); weights tuned on gallery + stairs only. Metric depth scored without alignment.



Corridor: refined depth (calibrated).

- Combined = refined when calibrated, focus estimate when rejected.
- Per scene it matches the better component; on the mean it beats both.

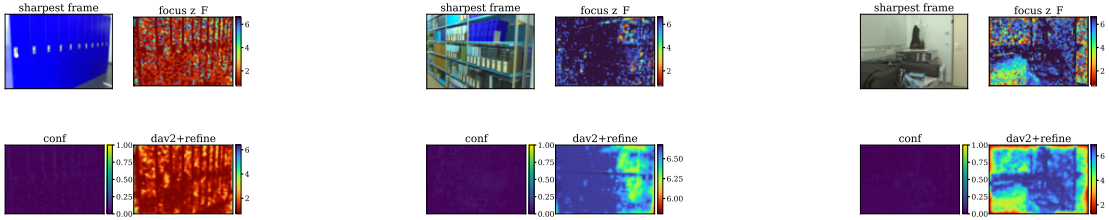
When the fit fails, say so



Slanted (calibrated = False): DA-V2 output anticorrelates with the focus estimate, $s \leq 0$, so the refinement is rejected and the combined output returns z_F.

- Scale fit alone on this scene: AbsRel 0.297 vs. focus estimate 0.087. The fallback is what makes the combined row win.
- Also exposed: defocus veiling lifts w near depth edges, and 2–3-frame stacks collapse the focus estimate entirely (AbsRel >10).

Real stacks: DDFF 12-Scene, qualitative only



Sharpest frame, focus estimate, confidence, DA-V2 + refinement on three DDFF test stacks.

- 120 Lytro stacks, no public ground truth (withheld for the benchmark server): qualitative evaluation only.
- Focus distances are not recorded; we rebuilt them from the paper's refocus geometry and verified ordering per stack.
- Real-stack confidence is low ($\bar{w} \approx 0.02$–0.05); per-scene confidence normalization gets calibration accepted on most stacks.

Takeaway, limitations, next

- **Benchmark:** relative DA-V2 is excellent after affine alignment (NYU δ_1 0.961) but the metric checkpoint (0.218) and median scaling (0.494) are not safe defaults.
- **Contribution:** focus gives absolute scale where texture exists; confidence-weighted fitting plus rejection turns it into a usable metric prior. Held-out AbsRel 0.206 beats DFF (0.233) and aligned DA-V2 (0.271).
- **Limitations:** texture dependence, veiling at occlusion edges, no DDFF ground truth, synthetic gap to real cameras.
- **Next:** more DDFF stacks, phone-captured focal stacks, per-checkpoint runtimes on the webpage.

github.com/DimitriChrysafis/cs566-project ▪ questions in the live Q&A